

Ch3 simple linear regression

Tuesday, September 22, 2026 9:31 AM

Recap:

$$Y = f(X) + \epsilon = (\beta_0 + \beta_1 X) + \epsilon$$

Assume β_0, β_1 are unknown parameters

Use sample (train data) to find a best fit

$$\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

where via optimization (calc)

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

to minimize

(residual sum of sqs)

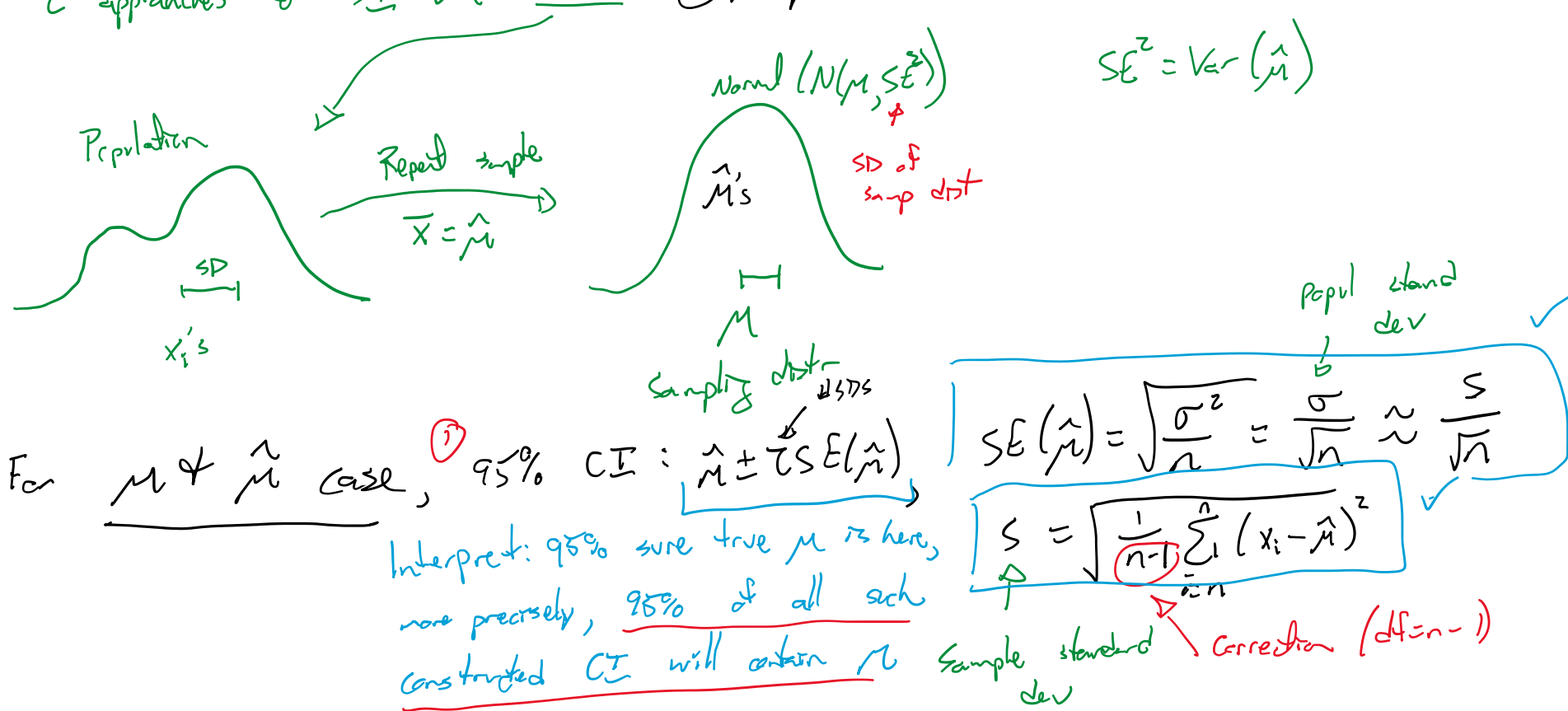
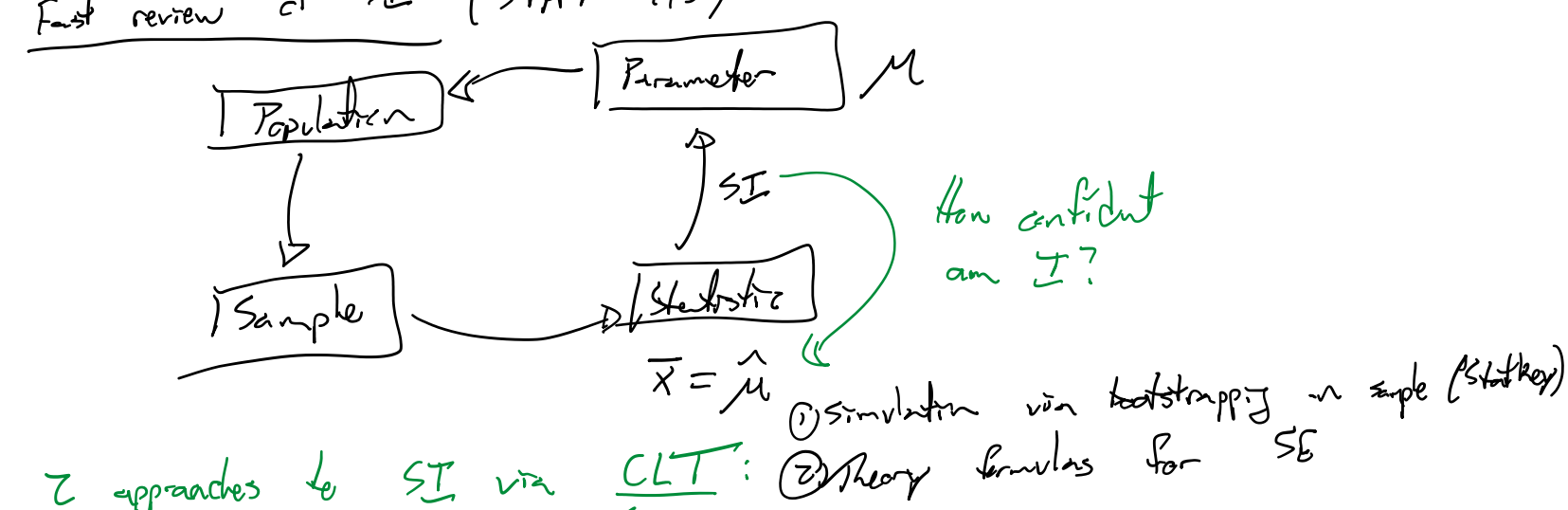
$$RSS = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

How reliable are estimates $\hat{\beta}_0 \approx \beta_0$ & $\hat{\beta}_1 \approx \beta_1$? (Statistical inference)

2 ways:

- 95% Confidence interval: $\hat{\beta}_i \pm 2SE$ (sample stats)
- Hypothesis test: $H_0: \beta_1 = 0$ (no X-Y rel)
 $H_a: \beta_1 \neq 0$ (X-Y rel)

Fast review of SI (STAT 145)



2) HT: $H_0: \mu = \mu_0$ (assume true) vs $H_a: \mu \neq \mu_0$ (tail)

Use z-score instead, $z = \frac{\hat{\mu} - \mu_0}{SE} \sim N(0, 1)$

prop of $\hat{\mu}$'s as or more extreme than $\hat{\mu}$ (p-val)

If p-val < 0.05 , reject H_0

Intend of a z-HT, often use a t-test (small samples better & same $N(0,1)$ for big samples)

(t-score) $t = \frac{\hat{\mu} - \mu_0}{SE} \sim t_{df}(0, 1)$ (default choice)

Apply to simple LR: $Y = f(X) + \epsilon = (\beta_0 + \beta_1 X) + \epsilon$

(sample stat) $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ (above)

95% CI: $\hat{\beta}_0 \pm 2SE(\hat{\beta}_0)$, $SE(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right)$

$\hat{\beta}_1 \pm 2SE(\hat{\beta}_1)$, $SE(\hat{\beta}_1) = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$

$\sigma^2 = \text{Var}(\epsilon)$ & $\hat{\sigma}^2 = \frac{1}{n-2} \sum (Y_i - \hat{Y}_i)^2$ (Residual SD)

HT: $H_0: \beta_1 = 0$ (X & Y not rel) vs $H_a: \beta_1 \neq 0$

(t-score) $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \sim t_{df}(0, 1)$, $df = n - 2$

Gives estimates & certainty of $\hat{\beta}_0, \hat{\beta}_1$, but what about $f(X) \approx \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ overall? 2 ways:

(old) $MSE = \frac{1}{n} \sum (Y_i - \hat{Y}_i)^2$ (y^2 units)

1) $RSE = \sqrt{\frac{1}{n-2} \sum (Y_i - \hat{Y}_i)^2}$ (e_i stand devs y unit)

(residual standard error) $0 \leq R^2 \leq 1$

2) R^2

$0 \leq R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} \leq 1$

(Residual SS) $RSS = \sum (Y_i - \hat{Y}_i)^2$

(Total SS) $TSS = \sum (Y_i - \bar{Y})^2$ (y_i variance)

In SLR: $R^2 = r^2 = \text{Corr}(X, Y)^2$ (correlation)

$= \left(\frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \sqrt{\sum (Y_i - \bar{Y})^2}} \right)^2$ (dot prod)

